Checking Quran Recitation from Noisy ASR Transcripts: A Competition with Agent-Assisted Annotation

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Abstract

A Quran-learning application must identify the ayahs in an ASR transcript and distinguish unresolved textual mistakes from repeats, repairs, and acceptable spelling differences. We propose two tasks: detecting and splitting ayahs, and labeling localized transcript/reference events. Task A has a public 258-recording benchmark. For Task B, we are building a private, human-reviewed set of 100 unique recordings: 40 recordings and 71 ayah chunks are approved so far. An agent track will annotate a separate unlabelled development pool and build an algorithm, whose final outputs are evaluated privately. We describe the current rubric and feasibility evidence, distinguish the earlier machine-labeled pilot from the new gold review, and identify public-input exposure that must be addressed before claiming an unseen-input competition test.

1 Task and motivation

Quran memorization commonly involves a listener who checks a learner's recitation. An application working from ASR must recognize opening formulas, repeated phrases, restarts, repairs, and missing or substituted words. A plain edit-distance comparison does not distinguish these cases. We assess observable transcript/reference differences; the labels do not establish whether the speaker or ASR caused them. Unwritten vowel and tajweed errors are outside scope.

In **Task A (detect and split)**, the input is a full transcript; the output is ayah identifiers and word assignments, or abstention on non-Quran input. The public v1.1 benchmark and scorecard already exist. In **Task B (event annotation)**, outputs have combined labels such as `substitution_mistake` and `repetition_benign`, with hypothesis and reference spans. Human review covers each entire recording, opening formulas, and all existing ayah splits. Before experiments, we will freeze whether executable algorithms receive full transcripts or supplied ayah chunks; full-transcript evaluation would report split quality separately.

The agent track extends algorithm development to *annotation*: entrants receive an unlabelled development pool, the Quran reference, definitions, and disjoint examples, then create annotations and an algorithm. The final executable is scored. Agreement with an agent's own development labels is not independent annotation accuracy. Good gold performance is evidence about the combined process, not proof that every development annotation is correct or that annotation caused an improvement.

2 Data and findings from human review

The public Task A release contains 258 transcripts with human-reviewed ayah assignments and splits: 254 Quranic recordings, four non-Quran negatives, and 1,267 chunks covering 604 ayahs from 45

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surahs. Its recording-level split has 151/107 rows and 711/556 chunks. This public split is not a secret competition test; separating recordings does not ensure different reciters. The inventory has 214 normalized recitation-text groups; duplicates must stay in one partition.

Current Task B review. We select for case coverage, preserving ASR text and comparing it with the clean Quran reference. As of 8 September 2026, **40 of 100 unique recordings** are approved: 71 chunks, 36 no-event chunks, 56 within-ayah events, and 16 opening-formula annotations. Sixty recordings remain to review. These purposive review counts do not estimate error prevalence. The rubric is versioned (v0.16); the final competition contract and metric are not frozen.

The review settled four rules. **Omissions** include missing beginnings and endings, with no benign exemption for a final ayah. **Repeats** span the original phrase and its extra copy against one reference copy. Wrong-then-correct attempts remain corrected events; a correct-then-incorrect restatement is `substitution_mistake` spanning both attempts. **Spelling** differences handled by agreed normalization receive no event; contextual residual forms use `spelling_benign`, with wasl explained in the note. This permits no blanket final-letter deletion. **Localization** uses verbatim words and half-open spans in original token arrays, including empty hypothesis spans for omissions. Merge continuous replacements, but preserve separate events across matching text or attempt boundaries.

Each event has one combined label; review status controls approval. Opening formulas and letter-name representations are retained. Possible ASR origin does not create an *uncertain* verdict, and repaired mistakes are not swallowed by benign repeats.

3 Evaluation and feasibility evidence

The existing Task A scalar sums detection, split, and abstention errors. Earlier work reported held-out scores of 0.760 for the incumbent and 0.079 for the best of six coding-agent runs.[1]

The legacy Task B pilot contains 99 machine-labeled chunks from 20 recordings, awaiting review under its old schema. Published pilot scores were 2.000 for an empty baseline and 1.111 for both naive diff and the incumbent on the public test subset. Those scores use separate type/verdict fields, an uncertainty category, and rules superseded by our review. They do not validate the new gold set. An audit also found 1,268 legacy Task B records versus 1,267 Task A chunks; that export mismatch is documented rather than silently repaired in released files.

For the new protocol, freeze matching, scalar weights, input contract, budgets, and hypotheses before runs. Report label and localization quality, missed mistakes, false flags on clean/benign/corrected cases, and per-label results. Matching must handle omission anchors and both-attempt spans, and prevent one broad prediction from earning credit for several distinct gold events. Re-run baselines under the frozen contract; no new Task B algorithm results are claimed.

4 Competition plan and exposure controls

Finish the 60 remaining reviews, audit duplicates and learner concentration, and prepare a separate unlabelled development bundle. Supply disjoint or synthetic examples and a fixed clarification exercise before measured runs. Agents receive no gold answers or gold-score feedback; final code is evaluated privately. Exclude reviewer files, source repositories, answer-bearing history, and annotation-session memory from agent runtimes.

Exposure limitation. All 40 approved inputs occur in the public Task A release; five overlap recordings with public legacy Task B pilot labels. New adjudications and selected membership remain private, but the inputs are not unpublished. Runtime isolation must exclude public-release retrieval and cannot itself establish absence of prior exposure. An unseen-input final competition ranking requires a separate collection of previously unreleased recordings, with reuse permission and de-identification verified before launch. That collection is planned, not already available.

The existing text-only release includes surrogate identifiers, a 26-recording sample, and checksums. New publication requires a privacy and reuse review. Shared documentation contains aggregate findings and synthetic examples; gold answers remain in a separate reviewer bundle.

[1] *Autoresearch with coding agents: generalizers and metric-maximizers on Quran recitation data*, AIST 2026. These are earlier Task A results, not Task B experiments.